

Recommendation Systems for Personalized Content Discovery

Participation Format

This is a team problem statement with a maximum of 02 participants per team.

Teams will work with the Netflix Prize Dataset to develop a personalized recommendation system capable of learning user preferences and generating relevant content recommendations.

1. Motivation

Every day, billions of users interact with recommendation systems that influence the movies they watch, the songs they listen to, the products they purchase, and the content they consume online.

Recommendation systems have become one of the most impactful applications of Artificial Intelligence and Machine Learning, powering personalization across platforms such as Netflix, Amazon, Spotify, YouTube, LinkedIn, and Instagram.

The ability to accurately understand user preferences and deliver relevant content directly influences user engagement, retention, and overall user experience. As a result, recommendation systems have evolved into a critical component of modern digital products.

One of the most influential milestones in the history of recommender systems was the Netflix Prize, a global machine learning competition launched to improve movie recommendation accuracy. The dataset released through this initiative remains one of the most widely studied datasets in recommendation system research and industry.

In this challenge, participants will step into the role of Machine Learning Engineers and Data Scientists tasked with building an intelligent recommendation engine capable of predicting user preferences and generating personalized content recommendations.

Dataset Overview

Dataset Link

<https://www.kaggle.com/datasets/netflix-inc/netflix-prize-data>

Participants will work with the Netflix Prize Dataset, one of the most influential datasets in the history of recommender systems.

Originally released as part of the Netflix Prize competition, the dataset was created to improve Netflix's movie recommendation algorithm and has since become a benchmark dataset for recommendation system research.

The dataset contains:

- 100,480,507 movie ratings
- 480,189 users
- 17,770 movies
- Ratings on a 1–5 star scale
- Rating timestamps for each interaction
- Movie metadata including movie IDs, titles, and release years

Each rating record consists of:

- User ID
- Movie ID
- Rating
- Date of Rating

The dataset is highly sparse, meaning that each user has rated only a small fraction of the available movies. This characteristic makes it particularly suitable for studying real-world recommendation challenges such as collaborative filtering, latent factor modeling, recommendation ranking, and cold-start problems.

Participants are encouraged to use appropriate subsets of the data if computational constraints prevent training on the full dataset.

2. Problem Statement

You are part of the machine learning team responsible for improving content discovery on a large-scale streaming platform.

Using historical user-item interaction data from the Netflix Prize Dataset, your task is to design and develop a recommendation system capable of delivering personalized content recommendations to users.

The recommendation engine should be capable of:

- Learning user preferences from historical interactions
- Predicting ratings for unseen content
- Generating personalized recommendations
- Identifying similarities between users and content items
- Improving content discovery through relevant recommendations

Participants may explore a variety of recommendation approaches, including but not limited to:

- User-Based Collaborative Filtering
- Item-Based Collaborative Filtering
- Matrix Factorization
- Singular Value Decomposition (SVD)
- Alternating Least Squares (ALS)
- Neural Collaborative Filtering
- Hybrid Recommendation Systems
- Ensemble Methods

The objective is not merely to predict ratings but to build a recommendation system that effectively captures user preferences and delivers meaningful recommendations.

3. Mandatory and Optional Tasks

Mandatory Tasks

A. Exploratory Data Analysis

Perform a detailed analysis of the dataset and identify:

- User activity patterns
- Content popularity trends
- Rating distributions
- Data sparsity characteristics

- Interesting observations and insights

Participants should clearly communicate the business and technical implications of their findings.

B. Recommendation Model Development

Develop at least one recommendation model capable of:

- Learning user preferences
- Predicting unseen ratings
- Generating personalized recommendations

Clearly explain the methodology used and justify its suitability for the problem.

C. Model Comparison

Implement and compare at least two recommendation approaches.

Possible comparisons may include:

- User-Based vs Item-Based Collaborative Filtering
- Matrix Factorization vs Collaborative Filtering
- Traditional Methods vs Deep Learning Approaches

Participants should discuss:

- Recommendation quality
 - Training complexity
 - Computational efficiency
 - Practical usability
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D. Recommendation Generation

Generate Top-K recommendations for users and analyze the recommendations produced.

Participants should demonstrate:

- Sample recommendations
- Recommendation quality

- Success cases
 - Failure cases
 - Key observations
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E. Evaluation

Participants must evaluate their recommendation system using the following **mandatory metrics**:

Mandatory Metrics

- **RMSE (Root Mean Squared Error)** – for measuring rating prediction accuracy.
- **MAP@10 (Mean Average Precision @ 10)** – for measuring recommendation ranking quality.

For the purpose of calculating **MAP@10**, a movie should be considered **relevant if its actual user rating is greater than or equal to 3.5**.

Participants must clearly describe:

- Their train-test split methodology
- The relevance definition used
- The procedure followed to generate Top-10 recommendations
- The methodology used to compute MAP@10

Additional Metrics (Optional)

Participants may additionally report:

- MAE
- Precision@K
- Recall@K
- NDCG
- Hit Rate
- Coverage
- Diversity Metrics

Participants must justify their evaluation strategy and discuss the trade-offs between rating prediction accuracy and recommendation ranking performance.

Optional Tasks

A. Explainable Recommendations

Provide explanations for recommendations generated by the model.

Example:

Users who enjoyed Movie A and Movie B were also likely to enjoy Movie C.

B. Cold Start Strategy

Discuss approaches for handling:

- New users
- New content items
- Sparse interaction histories

Implementation is optional.

C. Interactive Dashboard

Build an interface that allows users to:

- View recommendations
- Explore similar content
- Analyze recommendation scores

This task is optional and will not affect eligibility for evaluation.

D. Hybrid Recommendation System

Incorporate additional metadata or external information to improve recommendation quality.

E. Deployment

Deploy the recommendation system as:

- A Web Application
- A Dashboard

- An API Service

Deployment is optional.

4. Deliverables

Deliverable 1: Technical Report

Format: PDF

The report should include:

- Problem Understanding
- Exploratory Data Analysis
- Methodology
- Model Design
- Evaluation Metrics
- Experimental Results
- Recommendation Examples
- Key Insights
- Future Improvements

Maximum Length: 10 Pages

Deliverable 2: Source Code & GitHub Repository

The submission must include a **GitHub repository** containing:

- Data Processing Pipeline
- Model Training Pipeline
- Evaluation Scripts
- Recommendation Generation Module
- Documentation
- Instructions to reproduce results

The repository should be well-structured, documented, and reproducible.

Deliverable 3: Presentation

Format: PDF

Maximum Length: 8 Slides

The presentation should cover:

- Problem Overview
 - Approach
 - Key Insights
 - Experimental Results
 - Recommendation Examples
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5. Guidelines

1. Focus on Recommendation Quality

The objective is not simply to build a predictive model but to generate useful and relevant recommendations.

2. Justify Your Decisions

Clearly explain:

- Model choices
 - Evaluation metrics
 - Feature engineering decisions
 - Recommendation strategy
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3. Balance Performance and Simplicity

A simpler model with strong insights and thoughtful analysis is preferable to an unnecessarily complex solution.

4. Encourage Experimentation

Participants are encouraged to compare multiple approaches and explore creative techniques for improving recommendation quality.

5. Think Like a Machine Learning Engineer

Strong submissions will demonstrate not only model-building skills but also an understanding of user behavior, recommendation quality, ranking performance, and practical deployment considerations.

6. Prioritize Reproducibility

Your work should be easy to understand, reproduce, and extend. Proper documentation and repository organization are strongly encouraged.

Evaluation Criteria

Criterion	Weightage
Data Understanding & EDA	15%
Recommendation Model Development	30%
RMSE & MAP@10 Performance	20%
Recommendation Quality & Insights	20%

Innovation & Creativity 10%

Presentation & Documentation 5%

Data-Driven Investment Intelligence Using NIFTY-50 Market Data

Participation Format

This is a team problem statement.

Teams may consist of up to 02 participants.

Participants are expected to collaboratively design and develop an AI-powered investment intelligence platform using the provided NIFTY-50 dataset. All submitted work must be original and completed within the competition timeline.

The challenge evaluates not only machine learning and analytical capabilities but also the ability to transform financial data into practical decision-support systems that can assist investors in making informed decisions.

1. Motivation

Financial markets generate massive amounts of data every day, making it increasingly difficult for investors to identify meaningful patterns and make informed decisions.

While stock market data is publicly available, transforming raw historical prices and trading volumes into actionable insights requires sophisticated analytical techniques.

The NIFTY-50 index represents fifty of the largest and most influential companies listed on the National Stock Exchange of India, spanning multiple sectors such as Banking, Information Technology, Energy, Consumer Goods, Pharmaceuticals, Financial Services, and Manufacturing.

This challenge invites participants to build an AI-powered investment intelligence platform capable of extracting insights, identifying opportunities, assessing risks, and supporting data-driven investment decisions using historical market data.

Unlike traditional stock-price prediction competitions, the focus of this challenge is on creating practical decision-support systems that combine machine learning, financial analytics, explainability, and user-centric design.

Dataset Overview

Dataset Link:

<https://www.kaggle.com/datasets/rohanrao/nifty50-stock-market-data/data>

Additional datasets :

<https://www.kaggle.com/datasets/stoicstatic/india-stock-data-nse-1990-2020>

(Students are requested to use only the provided datasets)

Participants will work with the **NIFTY-50 Stock Market Dataset**, which contains historical market data for companies that have been part of the NIFTY-50 index of the National Stock Exchange (NSE) of India.

The dataset spans from **January 2000 to April 2021** and covers companies across multiple sectors including Banking, Information Technology, Energy, Consumer Goods, Pharmaceuticals, Financial Services, and Manufacturing.

The dataset consists of:

Historical Stock Data

Daily records for each company containing:

- Open Price
- High Price
- Low Price
- Close Price
- Traded Volume
- Turnover

Company Metadata

Additional information including:

- Company Name
- Sector

- Industry Classification

The dataset provides more than two decades of historical market activity, enabling participants to analyze long-term trends, market cycles, sectoral performance, volatility patterns, and investment opportunities.

Examples of derived indicators include:

- Moving Averages (MA)
- Exponential Moving Averages (EMA)
- Relative Strength Index (RSI)
- MACD
- Bollinger Bands
- Volatility Indicators
- Momentum Indicators

The challenge is designed to evaluate a participant's ability to transform raw historical market data into actionable investment intelligence and decision-support tools.

2. Problem Statement

You are part of the Data Science and Investment Analytics team of a financial technology company.

Your objective is to develop an intelligent investment intelligence platform capable of transforming historical market data into meaningful insights that can assist investors in making informed decisions.

The platform should help users:

- Analyze historical stock performance.
- Evaluate investment opportunities.
- Assess portfolio risk.
- Understand market trends and behavior.
- Generate actionable investment insights.
- Support data-driven decision making.

Participants are free to use any combination of:

- Machine Learning
- Time Series Forecasting
- Statistical Modeling

- Deep Learning
- Portfolio Optimization
- Financial Analytics

The final solution should go beyond simple stock-price forecasting and focus on delivering practical investment intelligence.

3. Mandatory and Optional Tasks

Mandatory Tasks

A. Stock Predictor Engine

Develop predictive models capable of forecasting future stock behavior using historical market data.

Examples:

- Predict future stock prices or returns over a specified investment horizon.
- Predict the direction of stock price movement (upward or downward).
- Forecasts should be supported by appropriate model evaluation and validation techniques.

Evaluation Metrics:

Participants may use metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), R^2 Score, and Directional Accuracy to assess model performance. Additional evaluation metrics may be incorporated if appropriately justified.

B. Portfolio Construction Module

Generate investment portfolios for different investor profiles.

Examples:

- Conservative Investor
- Balanced Investor
- Aggressive Investor

The system should recommend portfolio allocations and justify its decisions.

C. Risk Assessment Module

Evaluate the historical risk associated with individual stocks or portfolios.

Possible Metrics Include:

- Volatility
- Sharpe Ratio
- Sortino Ratio
- Maximum Drawdown
- Risk-Adjusted Return

Participants may explore additional risk metrics where appropriate.

Optional Tasks

A. Explainable AI Framework

Provide explanations supporting recommendations and insights generated by the system.

B. Personalized Investment Strategies

Design investment recommendations tailored to different risk tolerances and investment goals.

C. Market Anomaly Detection

Identify unusual patterns or events within historical market data.

Examples:

- Sudden Volatility Spikes
 - Extreme Drawdowns
 - Unusual Trading Activity
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D. Forecasting Module

Develop predictive models for:

- Future Price Trends
- Volatility Estimation
- Return Forecasting

Participants should clearly discuss assumptions and limitations.

E. Deployment

Deploy the solution as:

- A Web Application
 - An Interactive Dashboard
 - A Cloud-Based Service
 - A Local Desktop Application
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4. Constraints

Participants may use only the datasets provided by the organizers.

Allowed

- Feature Engineering
- Statistical Transformations
- Technical Indicators
- Machine Learning Models
- Deep Learning Models
- Portfolio Optimization Techniques

Examples include:

- Moving Averages
- Exponential Moving Averages
- RSI
- MACD
- Bollinger Bands
- Momentum Indicators
- Volatility Indicators

Not Allowed

- Live Market Data
- Financial APIs
- News Datasets
- Social Media Sentiment Data

- Proprietary Financial Data
- Alternative Market Datasets

The objective is to evaluate how effectively participants can leverage the provided dataset.

5. Deliverables

Deliverable 1: Working Prototype

A functional application demonstrating the implemented capabilities.

Deliverable 2: Technical Report

Format: PDF

The report should include:

- Exploratory Data Analysis (EDA)
- Feature Engineering
- Methodology
- Model Architecture
- Portfolio Construction Logic
- Risk Assessment Methodology
- Explainability Techniques
- Key Insights

Maximum Length: 12 Pages

Deliverable 3: Public GitHub Repository

The repository should contain:

- Source Code
- Model Files
- Configuration Files
- Documentation

The solution should be reproducible.

Deliverable 4: README.md

Include instructions for:

- Environment Setup
 - Dependency Installation
 - Running the Application
 - Reproducing Results
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6. Guidelines

1. Focus on Decision Support

The objective is not simply to predict stock prices but to assist users in making informed investment decisions.

2. Justify Your Recommendations

All investment recommendations and portfolio suggestions should be supported by quantitative evidence.

3. Emphasize Explainability

Users should be able to understand why a recommendation or insight was generated.

4. Balance Complexity and Usability

Sophisticated analytics should be presented in a manner that remains understandable and actionable.

5. Encourage Innovation

Participants are encouraged to explore novel approaches for investment intelligence, risk assessment, portfolio optimization, and explainable AI.

7. Evaluation Criteria

Criterion	Weightage
Quality of Investment Insights	25%
Technical Innovation	20%
Portfolio & Risk Analytics	20%
Explainability & Transparency	20%
Reproducibility & Documentation	15%
